

Orders Solutions

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§1 Walkthrough Problems

Problem 1.1. Find all integers $n \geq 1$ such that n divides $2^n - 1$.

Proof. We show that the answer is $n = 1$ only. For the sake of contradiction, assume that some $n > 1$ works. Throughout this proof, p represents an arbitrary prime number. Consider now any $p \mid n$.

Claim 1.2 — $p \neq 2$

Proof. Obvious, as $p = 2 \implies 2 \mid n \mid 2^n - 1$ which is a contradiction. $\square$

Claim 1.3 — If $p \mid n$, then $\text{ord}_p 2 \mid \gcd(n, p - 1)$.

Proof. Since $p \mid n \mid 2^n - 1$, we get $2^n \equiv 1 \pmod{p}$ and so $\text{ord}_p 2 \mid n$. Moreover, Fermat gives $2^{p-1} \equiv 1 \pmod{p}$, so $\text{ord}_p 2 \mid p - 1$ and the desired result follows. $\square$

Claim 1.4 — For any $n \in \mathbb{Z}_{\geq 2}$, there exists a $p \mid n$ such that $\gcd(n, p - 1) = 1$.

Proof. Choosing the smallest $p \mid n$ works (where p can be 2, for even n). $\square$

Claim 1.5 — $\text{ord}_p 2 = 1$

Proof. Just choose the smallest $p \mid n$ which forces this. $\square$

All of the above gives us that $2 \equiv 1 \pmod{p}$, which is a contradiction. Thus, $n = 1$ is the only solution. $\square$